Chocolate at heart: the anti-inflammatory impact of cocoa flavanols.

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Chronic and acute inflammation underlies the molecular basis of atherosclerosis. Cocoa-based products are among the richest functional foods based upon flavanols and their influence on the inflammatory pathway, as demonstrated by several in vitro or ex vivo studies. Indeed, flavanols modify the production of pro-inflammatory cytokines, the synthesis of eicosanoids, the activation of platelets, and nitric oxide-mediated mechanisms. A relative paucity of data still characterizes the in vivo implications of these findings albeit there have been studies suggesting that the regular or occasional consumption of cocoa-rich compounds exerts beneficial effects on blood pressure, insulin resistance, vascular damage, and oxidative stress. Accordingly, rigorous controlled human studies with adequate follow-up and with the use of critical dietary questionnaires are needed to determine the effects of flavanols on the major endpoints of cardiovascular health.